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A REFINED APPARENT POLAR WANDER CURVE FOR THE TRANSDANUBIAN CENTRAL MOUNTAINS AND ITS BEARING ON THE MEDITERRANEAN TECTONIC HISTORY *

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ABSTRACT

Márton, E. and Márton, P., 1981. A refined apparent polar wander curve for the Transdanubian Central Mountains and its bearing on the Mediterranean tectonic history. In: E. McClelland Brown and J. VandenBerg (Editors), Palaeomagnetism of Orogenic Belts. *Tectonophysics*, 98: 43–57.

Recently obtained palaeomagnetic data have given a fairly detailed apparent polar wander (A.P.W.) curve for the Mesozoic of the Transdanubian Central Mountains which has been compared with other A.P.W. curves from the Mediterranean area. The similarity of the A.P.W. curves enabled a unified A.P.W. curve to be constructed for the Central Mediterranean region. It has been shown that the new Mediterranean A.P.W. curve is in full agreement, even in details, with the updated A.P.W. curve for Africa.

This confirms the existence during the entire Mesozoic of a single Central Mediterranean megatectonic unit rigidly attached to Africa. Its positions with respect to the pole can now be traced back in more detail than before.

INTRODUCTION

In a recent paper Márton and Márton (1981) have shown that the distribution of the Mesozoic palaeomagnetic poles for the Transdanubian Central Mountains displays an "African" apparent polar wander pattern (e.g. VandenBerg, 1979). With respect to this pattern, the single Tithonian pole seemed to be anomalous as it differed significantly from the main cluster of poles.

The discrepant pole encouraged further palaeomagnetic research as it implied that the movement of the Transdanubian Central Mountains might have been larger

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relative to the magnetic pole than the magnitude indicated by the Mesozoic poles other than the Tithonian.

In what follows, we summarize, for the convenience of the reader, all former and new palaeomagnetic results from the Transdanubian Central Mountains, and discuss their implications for the Mesozoic tectonic history of the Mediterranean area and other fragments of Gondwanaland.

MESOZOIC AND PALAEOGENE PALAEOMAGNETIC RESULTS FROM THE TRANSDANUBIAN CENTRAL MOUNTAINS

A simplified geological map of the Transdanubian Central Mountains with the palaeomagnetic sampling localities is given in Fig. 1. The outcrops of the Mesozoic formations of the same age form southwest–northeast trending zones. In the southeast, Palaeozoic and Triassic rocks occur. The former includes a granite area with outcrops of Eocene andesite. The median zone is occupied by Jurassic and Cretaceous formations while the narrow northeast strip consists of Triassic rocks. Structurally, these formations are believed to form an autochthonous asymmetrical synclinorium that folded before the Senonian (Fülöp, 1968).

Sixty-four localities have been sampled so far from the Mesozoic and Palaeogene. The samples were measured on JR-4 spinner magnetometers with an r.m.s. noise level of $4 \cdot 10^{-6}$ A/m. Seven andesite and thirty-three sediment collections yielded useful palaeomagnetic results (Fig. 1 and Table I), whereas the remainder were too weak for precise measurements.

On calculating the locality means in Table I, some of the samples had to be disregarded either due to low intensity as in localities Nos, 2, 4, 7, 8, 29, and 33, or overprinted magnetization, which means that samples with imperfectly removed overprint were rejected and the locality mean was based on apparently single-component remanence as in localities 6, 15, 16, 22, 24, 25, 27, 28 and 30.

The diagrams which follow are examples of how the different rocks responded to magnetic cleaning. Orthogonal projections of the successive end points of the magnetization vector during progressive alternating field (AF) and thermal demagnetization are shown in Fig. 2, intensity and susceptibility changes on thermal treatment in Fig. 3.

The Eocene andesite (loc. 1, Table I) was cleaned in an alternating field. The demagnetization curve displays a single component from 0.01 T on (Fig. 2a).

A red marl specimen (loc. 6, Table I) carries an overprint in the direction of the present field (Fig. 2b), but it is practically removed at 525°C. The susceptibility vs. temperature curve (Fig. 3a) reveals that the overprinting magnetization resides in maghemite which converts on heating to haematite with a lower susceptibility.

Usually, the natural remanence of the grey carbonates cleans readily. Such an example is shown in Fig. 2c. (loc. 19., Table I). A single component is indicated from 400°C on. The stability of the magnetic minerals on heating is remarkable (Fig. 3b).

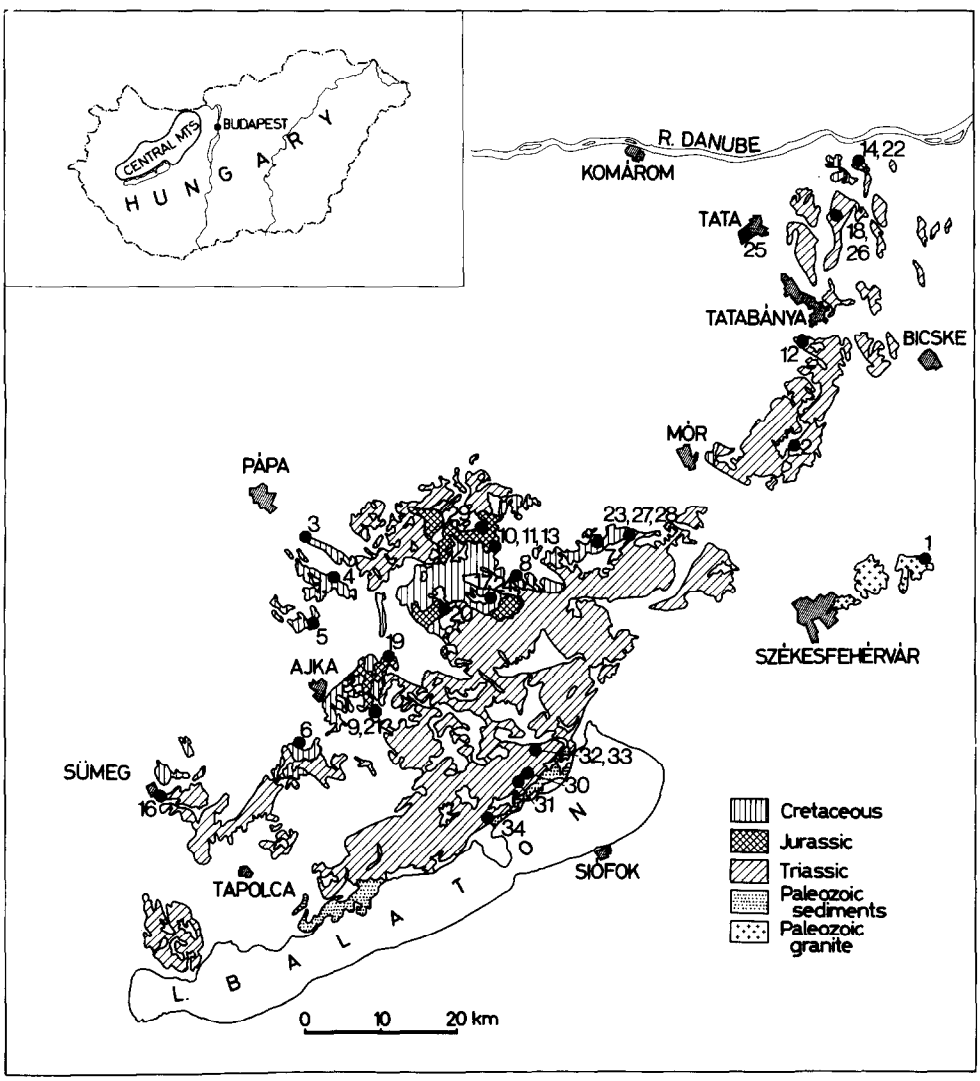


Fig. 1. Geological sketch map of the Transdanubian Central Mountains. Numbers indicate the palaeomagnetic localities.

TABLE I

Summary of the palaeomagnetic results from Mesozoic and Palaeogene rocks of the Transdanubian Central Mountains. Mean geographic coordinates are: lat. 47.5°N, long. 18.1°E. All computations were carried out following Fisher (1953).

	Age	Rock type	Locality	N/N_o	Magnetization before tilt correction			Magnetization after tilt correction			Cleaning Thermal (°C)
					Decl.°	Incl.°	α_{95}°	Decl.°	Incl.°	α_{95}°	AF (Tesla)
	EOCENE										
1.	Upper Eocene	andesite	Velencei Mts.	7*	329	47	10	329	47	10	0.02-0.1T
2.	Upper Eocene	grey marl	Gánt	3/6	137	-60	14	137	-60	14	500
	CRETACEOUS										
3.	Senonian	grey marl	Tapolcafő	11/12	145	-57	8	132	-53	8	150-300
4.	Senonian	grey marl	Bakonyjákó	8/13	347	59	11	330	57	12	150-200
5.	Senonian	grey marl	Magyarpolány	5/6	314	72	14	320	52	9	500
6.	Senonian	red marl plus bauxite	Halimba	10/13	305	58	11	312	50	11	550
7.	Albian	grey limestone	Jásd	22/54	284	47	10	292	40	10	300-565
8.	Albian	grey limestone	Olaszfalu	9/28	297	25	16	300	44	16	250-500
9.	Albian	grey limestone	Urkut	8/10	309	60	16	308	30	16	525-550
10.	Aptian	grey limestone	Borzavár I.	11/11	290	42	7	294	47	7	500
11.	Aptian	grey limestone	Borzavár II.	6/9	271	42	10	275	43	10	500-525
12.	Aptian	grey limestone	Vértessomlyó	7/7	289	27	16	285	47	10	500
13.	Valanginian-Hauterivian	grey limestone	Borzavár II.	8/10	258	40	9	263	42	9	525-625
14.	Valanginian-Hauterivian	grey marl	Lábatlan	10/14	299	45	10	290	39	10	400-500
15.	Berriasian	grey limestone	Sümeg	56/142	312	-42	5	277	31	5	500-545

	JURASSIC									
16.	Tithonian	grey limestone	Sümeg	42/58	329	-36	7	260	32	7	500-545
17.	Tithonian	grey limestone	Lókut	14/15	249	37	4	264	38	4	500
18.	Tithonian	grey limestone	Tardos	11/11	267	17	7	278	42	7	560
19.	Tithonian	grey limestone	Városlőd	6/6	91	42	8	256	52	8	520-550
20.	Bajocian-Bathonian	red limestone	Hárskut	30/30	304	48	9	307	39	9	500-600
21.	Bajocian	grey limestone	Urkut	10/10	300	40	33	314	37	17	400-540
22.	Toarcian-Bajocian	red limestone	Lábatlan	53/120	308	46	4	293	44	4	450
23.	Toarcian	red limestone	Bakonycsernye I.	15/15	279	53	6	295	52	6	450
24.	Lower Jurassic	red limestone	Lókut	21/28	279	53	8	288	47	8	450
25.	Lower Jurassic	red limestone	Tata	10/13	317	57	6	332	58	6	500
26.	Pliensbachian	red limestone	Tardos	6/6	312	58	11	310	57	11	400
27.	Pliensbachian	red limestone	Bakonycsernye I.	51/152	281	57	3	299	55	3	450
28.	Pliensbachian	red limestone	Bakonycsernye II.	7/13	291	56	9	302	48	9	450
	TRIASSIC									
29.	Norian-Raethian	grey limestone	Borzavár	4/6	308	42	9	306	50	9	400
30.	Ladinian-Carnian	grey limestone	Csopak I.	9/13	328	66	4	314	33	4	500
31.	Ladinian-Carnian	grey limestone	Csopak II.	5/6	302	50	24	303	35	24	200
32.	Ladinian	grey limestone	Felsőörs	5/8	310	58	14	291	35	14	400-500
33.	Anisian	grey limestone	Felsőörs	33/95	306	62	7	304	41	7	400-500
34.	Werfenian	grey limestone	Balatonfüred	12/13	320	34	5	315	38	5	500

N/N_o number of samples used/collected

* number of localities

References: 1, 2, 6, 7, 8, 9, 19, 21, 32, 33: Márton, E., unpublished
3, 4, 10, 13: Márton, E. and Veljovic, D., unpublished
5, 11, 12, 14, 17, 23, 24, 25, 26, 27, 28, 29, 30, 31, 34: Márton, E. and Márton, P., 1981
15, 16: Márton, E., 1982
18, 20, 22: Márton, P. and Márton, E., unpublished

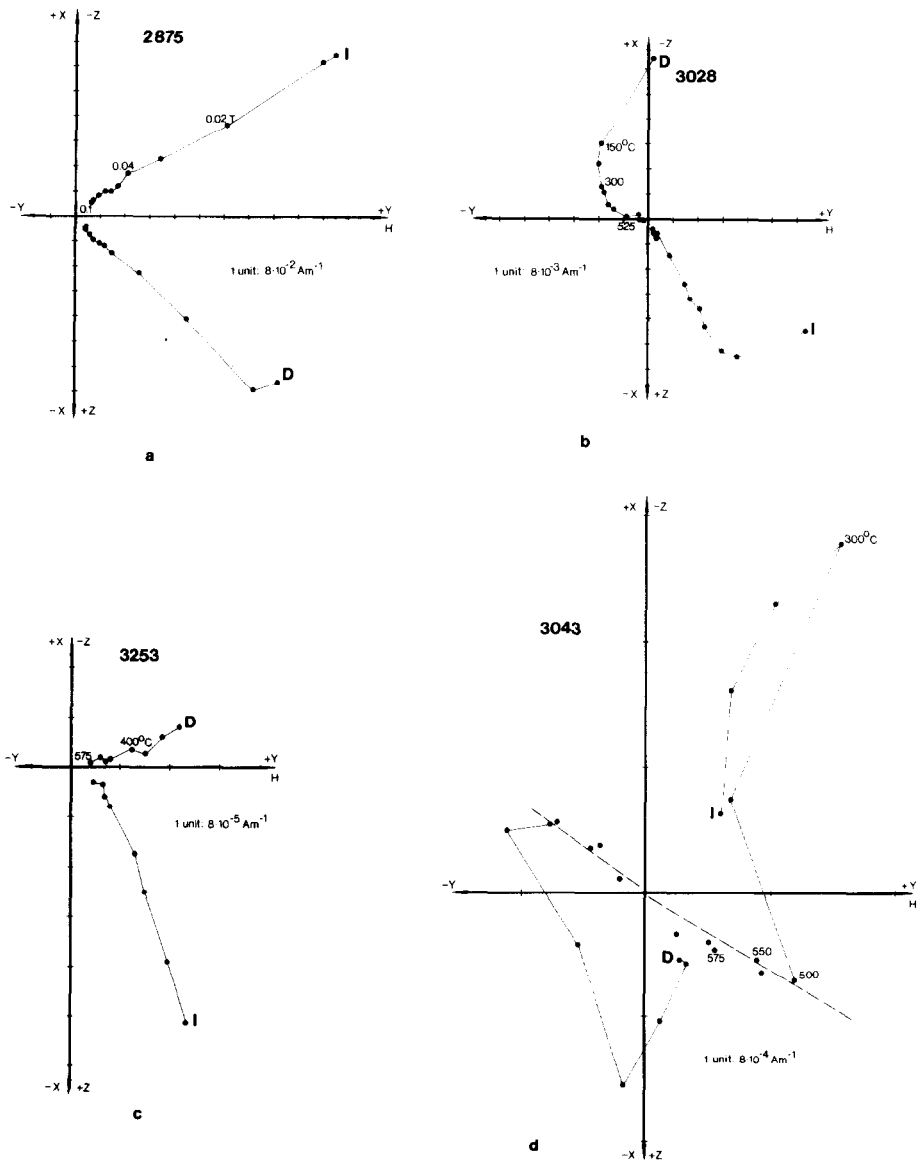


Fig. 2. Orthogonal projections of the successive end points of the magnetization vector during progressive demagnetization. a. Eocene andesite, AF cleaned. b. Senonian red marl. c. Tithonian grey limestone. d. Bathonian red limestone. b, c and d thermally demagnetized.

The magnetizations of the red limestones are the most complicated of all (loc. 20, Table I). The component with the highest blocking temperature is overprinted twice. Drastic changes in the direction occur at 300°C and 500°C, respectively (Fig. 2d).

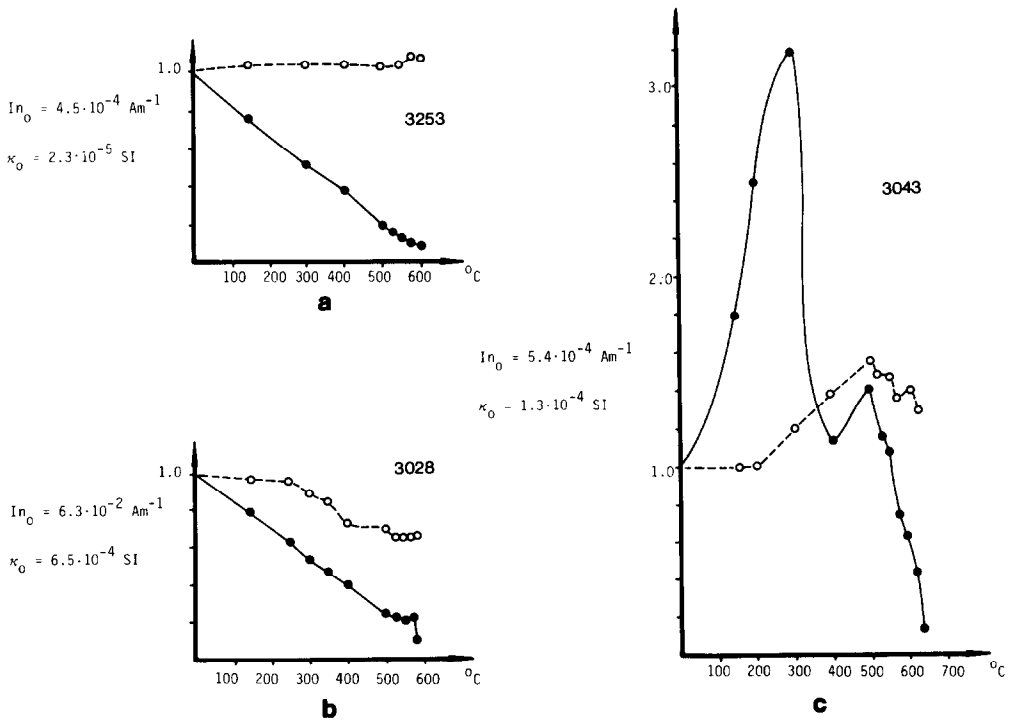


Fig. 3. Intensity and susceptibility changes of pilot specimens during thermal treatment. a. Senonian red marl. b. Tithonian grey limestone. c. Bathonian red limestone. Solid symbols: intensity, open symbols: susceptibility, both normalized at the values in the natural state.

The first change is due to the removal of the magnetization acquired in the present field, the second to the removal of an ancient magnetization antiparallel to the most stable component (Fig. 4). The increase in susceptibility at relatively low temperatures can be explained by the conversion of hydrous iron oxides to haematite and perhaps maghemite, too, whereas the decrease in susceptibility above 500°C can be explained by the reorganization of the newly formed maghemite to haematite (Fig. 3c). When the overprints are removed, the magnetization becomes single-component above temperatures of 500°–525°C (Figs. 2d and 4).

Based on the locality means, average declinations and inclinations were calculated for stages or groups of stages in the Palaeogene and the Mesozoic, before and after tilt correction, and palaeomagnetic poles computed from the corrected directions (Table II). Correction for local bedding improves k and α_{95} in general, but the efficiency of the tilt correction (on the statistical parameters) is strongly dependent on the variation of the local bedding attitude within one group: when the dip angles are small, tilt correction has only a slight influence (e.g. Norian–Pliensbachian);

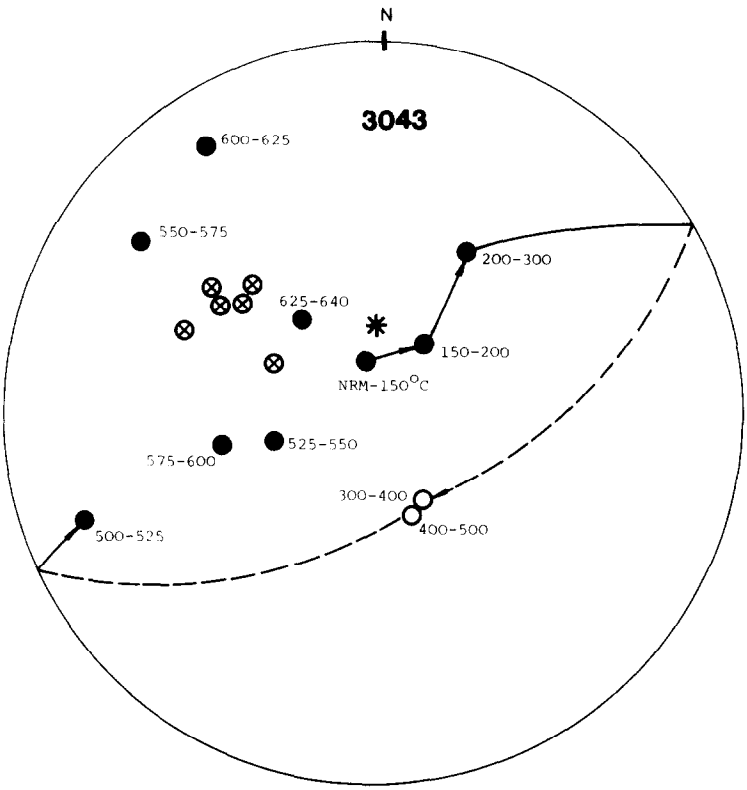


Fig. 4. Stereographic plot of the directions of the magnetizations removed during progressive thermal demagnetization (closed symbols: lower hemisphere, open symbols: upper hemisphere) and the end points of the cleaning at 500°–625°C (crosses encircled) in a bathonian red limestone sample. Star: present field.

when the bedding attitudes are different, the improvement on correction can be striking (e.g. Tithonian).

Apart from the stability of the magnetization on laboratory treatment, the positive response to the fold test in most age groups and/or the regional consistency of the coeval palaeomagnetic directions serve as additional evidence in favour of the primary nature of the characteristic magnetization.

The palaeomagnetic results summarized in Table I confirm the pole position for the Tithonian and permit us to construct a more refined apparent polar wander (A.P.W.) curve for the Mesozoic (Fig. 5). The A.P.W. curve exhibits the "African loop" as observed before (Márton and Márton, 1981) but this is more elongated now and has a hairpin bend at Late Jurassic–Early Cretaceous time.

TABLE II

Mean palaeomagnetic directions and palaeomagnetic poles (lat. N, long. E) for different age groups from the Transdanubian Central Mountains

Age	N_o	N	Magnetization before tilt correction				Magnetization after tilt correction				Palaeomagnetic pole		Tilt correction
			Decl.°	Incl.°	k	α_{95}	Decl.°	Incl.°	k	α_{95}°	Lat.°	Long.°	locality (cf. Table I): azimuth/dip
Eocene	7+6*	7+1=2*	324	54	-	-	324	54	-	-	275	60	1: 0/0, 2: 0/0
Senonian	12	4	324	62	52	13	318	53	195	7	280	56	3: 253/10, 4: ~, 5: ~, 6: 330/10
Albian	4	3	296	44	17	30	300	38	71	15	281	36	7: 336/10, 8: 102/20, 9: ~
Aptian	5	3	284	37	46	18	284	46	132	11	298	29	10: 76/6, 11: 10/6, 12: ~
Berriasian-Hauterivian	3	3	290	10	2	112	277	38	46	18	298	20	13: 10/6, 14: 234/12, 15: 293/104
Tithonian	4	4	277	-7	3	69	265	41	55	12	308	14	16: 293/104, 17: 353/19, 18: 54/37, 19: 135/120
Toarcian-Bathonian	7	4	298	47	66	11	304	43	59	12	281	41	20: 324/9, 21: ~, 22: 217/15, 23: 12/12
Norian-Pliensbachian	10	6	298	55	52	9	305	53	68	8	290	47	24: 335/10, 25: 83/10, 26: 262/2, 27: 12/12, 28: 344/12, 29: 140/8
Werfenian-Carnian	6	5	313	54	34	13	305	37	91	8	276	39	30: 302/35, 31: 308/15, 32: 279/33, 33: 279/33, 34: 200/8

N_o number of localities sampled

N number of useful localities

* andesites + sediments

~ different positions of samples, improved grouping on tilt correction (see Table I.)

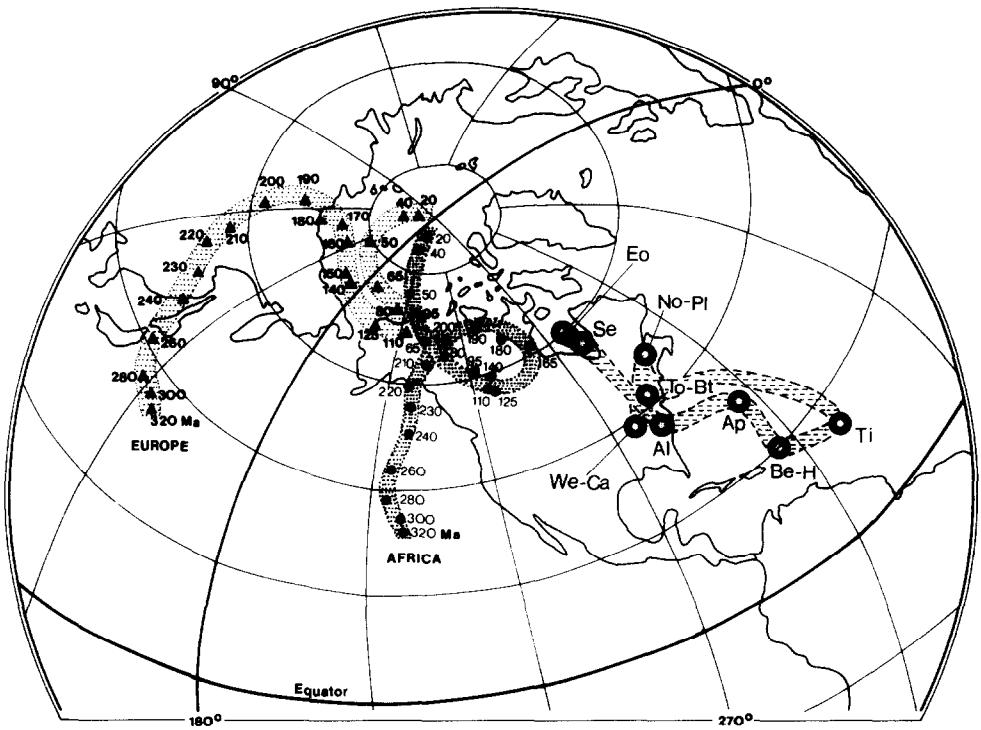


Fig. 5. Mesozoic and Palaeogene palaeomagnetic poles for the Transdanubian Central Mountains. The "pseudo-African" and "pseudo-European" polar wander curves are shown for comparison. Both are derived from the North American A.P.W. curve (Irving, 1977) using the stage poles for the opening of the South Atlantic (Sclater et al, 1977). *E0* = Eocene, *Se* = Senonian, *Al* = Albian, *Ap* = Aptian, *Be-H* = Berriasian–Hauterivian, *Ti* = Tithonian, *To-Bt* = Toarcian–Bathonian, *No-Pl* = Norian–Pliensbachian, *We-Ca* = Werfenian–Carnian.

DISCUSSION

Certain tectonic units in the Mediterranean are known to have Mesozoic and Palaeogene palaeomagnetic poles that are similar to the African poles—Southern Alps, Italy (Lowrie et al., 1980; VandenBerg and Wonders, 1980)—or can be made "African" after correcting for a late Tertiary rotation—Umbria, Italy (VandenBerg, 1979), Transdanubian Central Mountains, Hungary (Márton and Márton, 1981) Istria peninsula, Yugoslavia (Márton, E. and Veljovic, 1983), Apulia, Italy (VandenBerg, 1982).

Palaeomagnetic results from Sardinia (Edel et al., 1981; Montigny et al., 1981; Horner and Lowrie, 1981) and the West Carpathians of Czechoslovakia (Krs et al., 1979; Koráb et al., 1981) may be interpreted as rotated African poles too (Márton,

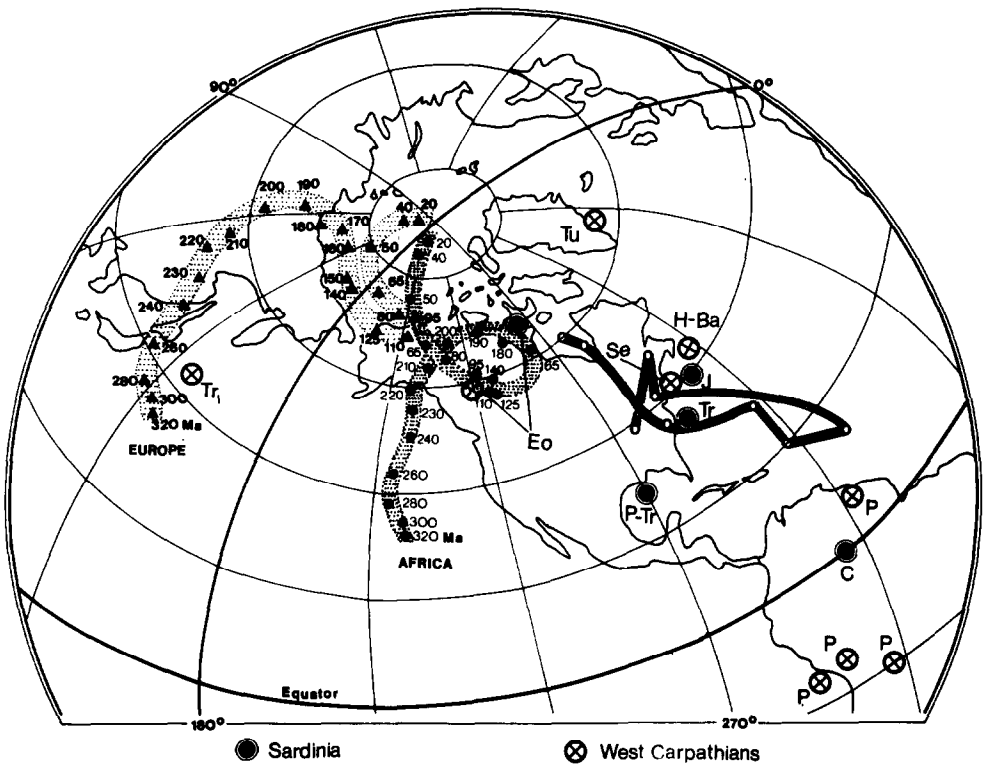


Fig. 6. Palaeomagnetic poles from the West Carpathians and Sardinia. The A.P.W. curve for the Transdanubian Central Mountains (ages: cf. Fig. 5.) and the "pseudo-African" and "pseudo-European" polar paths (cf. Fig. 5.) are shown for comparison. *Eo* = Eocene, *Se* = Senonian, *Tu* = Turonian, *H-Ba* = Hauterivian–Barremian, *J* = Jurassic, *Tr* = Triassic, *T-P* = Permo-Triassic, *P* = Permian, *C* = Carboniferous.

E. 1982). The direction and amount of rotation with respect to Africa is similar in all cases, but the pattern of the apparent polar wander is not clear in the West Carpathians and data are relatively scarce from Sardinia (Fig. 6).

If we compare the published segments of the Mesozoic and Palaeogne A.P.W. curves from the Southern Alps, Umbria, Istria peninsula and the Transdanubian Central Mountains, we find a common trend in the apparent polar movements. With the exception of the Southern Alps, the poles of the same age agree quite well indicating that Umbria, Istria peninsula and the Transdanubian Central Mountains have not participated in a major relative movement. Consequently, they can be characterized by a common A.P.W. curve. This curve includes an "unusual" Barremian pole form Umbria (Lowrie et al., 1980), which is anomalous in comparison with the "pseudo-African" polar path in the same sense as the Transdanubian poles for the Tithonian and the Early Cretaceous.

Supporting evidence for the existence of "unusual" pole positions in Umbria comes from a yet unpublished result for a Late Jurassic–Early Cretaceous pole of lat. 18°N, long. 288°E based on 100 samples from three localities (Cirilli et al., 1983).

Table III summarizes the mean palaeomagnetic poles used in constructing the common A.P.W. path for Umbria, the Transdanubian Central Mountains and Istria peninsula (Central Mediterranean region).

In order to demonstrate the similarity of the A.P.W. curves for the Central Mediterranean region and Africa, we relocated the common A.P.W. curve for the Central Mediterranean region by a 35° clockwise rotation about an arbitrary rotation pole of lat. 47.5°N, long. 18°E (Fig. 7).

It can be seen from Fig. 7 that the A.P.W. for the Central Mediterranean region is a well defined curve (though not in statistical terms, as a maximum three units contribute to a combined pole, cf. Table III). The curve undoubtedly displays the characteristic clockwise loop which agrees closely with the inferred polar wandering for Africa, while major deviations are indicated for the Late Jurassic–Early Creta-

TABLE III

Mean stage poles for the Central Mediterranean region (lat. N, long. E)

Geological stage/period	Pole from			Mean palaeomagnetic pole			
	TCMts*	Umbria	Istria	Lat.°	Long.°	k	A_{95}°
Eocene	x	x		63	266	120	23
Senonian	x	x	x	61	272	43	20
Turonian		x	x	49	271	365	13
Cenomanian		x	x	45	285	78	29
Albian	x	x	x	42	277	61	16
Aptian	x	x	x	37	285	42	19
Barremian-Valanginian	x	x	x	27	293	26	25
Tithonian-Berriasian	x	x		19	297	73	30
Kimmeridgian-Middle Jurassic	x	x		39	287	103	25
Pliensbachian-Late Jurassic	x	x		47	295	283	15
Early-Middle Triassic	x	x		38	266	56	34

* Transdanubian Central Mountains

References: Transdanubian Central Mountains: Table I, this paper; Umbria: Channell et al. (1978), Lowrie et al. (1980), VandenBerg, (1979); Istria: Márton, E. and Veljovic (1983)

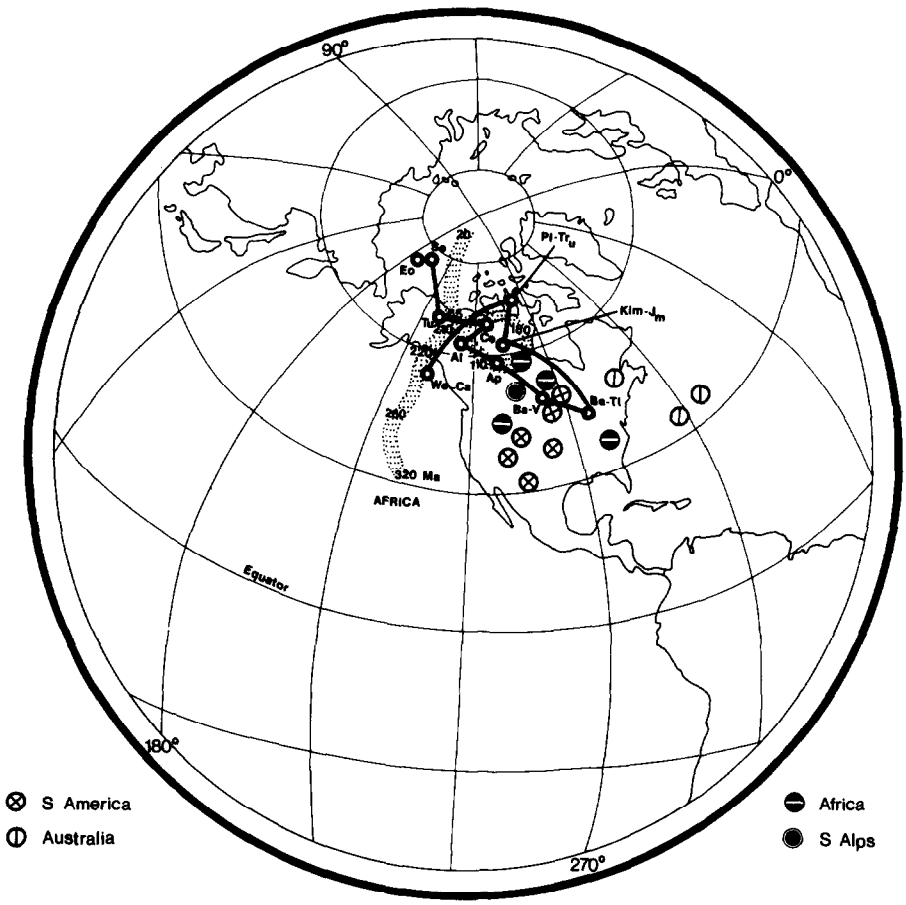


Fig. 7. The common A.P.W. curve for the Central Mediterranean relocated by a 35° clockwise rotation about an arbitrary rotation pole of lat. 47.5°N, long. 18°E. The "pseudo-African" A.P.W. curve and discrepant poles from Africa, the Southern Alps, South America and Australia (the latter two corrected for the relative movement with respect to Africa) are shown for comparison. The Australian poles (Schmidt and Embleton, 1982) are of Early–Late Jurassic, the African and South American poles (Schult et al., 1981) of Early Cretaceous age. The pole from the Southern Alp is Barremian (Lowrie et al., 1980). *Eo* = Eocene, *Se* = Senonian, *Tu* = Turonian, *Ce* = Cenomanian, *Al* = Albian, *Ap* = Aptian, *Ba-V* = Barremian–Valanginian, *Be-Ti* = Berriasian–Tithonian, *Kim-J$_m$* = Kimmeridgian–Middle Jurassic, *Pl-Tr$_u$* = Pliensbachian–Late Triassic, *We-Ca* = Carnian–Werfenian.

ceous. The new element is a very fast polar movement—away from the main cluster of the Mesozoic poles in the Late Jurassic and back to it in the Early Cretaceous.

In Fig. 7, the relocated curve is compared with the "pseudo-African" polar path and some other relevant palaeomagnetic observations from Africa, South America—the latter corrected for the initial opening of the South Atlantic (Schult et al., 1981)—and the Southern Alps (Lowrie et al, 1980).

Schmidt and Embleton (1982) recently called attention to the mobility indicated by some of the Mesozoic poles for Australia. The discrepant palaeomagnetic poles deviating from the cluster of the Mesozoic poles for Africa—represented by the "pseudo-African" polar path—are not very well defined, and the grouping of the poles from the different fragments of Gondwanaland is also poor. Despite these deficiencies, the palaeomagnetic data strongly suggest a uniformly larger polar wandering for the Gondwana fragments than would be expected from the "pseudo-African" A.P.W. curve.

CONCLUSIONS

The similarity of the palaeomagnetic polar wanderings as well as the individual poles of the same age group for the Transdanubian Central Mountains, Istria peninsula and Umbria indicate that these regions formed a single tectonic unit, the Central Mediterranean region, during the Mesozoic. Probably, this major tectonic unit was of wider extent than the above mentioned three areas.

By averaging the palaeomagnetic poles of the same geological stage(s) we have constructed a common A.P.W. curve for the Central Mediterranean region which essentially consists of a clockwise loop from the Early Triassic through the Palaeogene, similarly to the "pseudo-African" A.P.W. curve. When the former is rotated with respect to the latter for comparison, its Late Jurassic–Early Cretaceous segment turns out to be more elongated due south than the respective part of the inferred curve.

The discrepant Jurassic–Early Cretaceous palaeomagnetic poles from the southern continents are features we identify with the more elongated loop of the A.P.W. curve for the Central Mediterranean region.

Since the age and, more importantly, the chronological succession of the palaeomagnetic poles in the Mediterranean is better controlled than on the southern continents, we use the Mediterranean results to date the characteristic changes in the Mesozoic polar wandering of Gondwana. We suggest that rapid polar movement commenced in the Late Jurassic. The pole attained the extreme position in the Tithonian and the movement decelerated in the Aptian.

It has been customary over the last years to compare measured palaeomagnetic poles of the Mediterranean region with the "pseudo-African" A.P.W. curve derived from the smoothed North American polar path. However, during the Late Jurassic–Early Cretaceous period of fast polar movement, neither this curve nor the one derived by rotating the measured North American poles to Africa display the enlarged polar wandering of the Mediterranean region and the southern continents. Nevertheless, it can be shown that the Late Jurassic–Early Cretaceous segment of the transferred North American A.P.W. curve becomes fairly similar to the corresponding part of the Mediterranean A.P.W. path if the former is generated by keeping the North Atlantic closed before 120 Ma ago. Subsequently a fast sea-floor

spreading is to be assumed to account for the known amount of relative displacement between North America and Africa by about 100 Ma ago.

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